

## INHERITANCE GENETICS

# Module map

- Module 09: Inheritance Genetics
- Connected lab: lab-09\_inheritance-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 09

### Inheritance Genetics

1

Alleles and  
genotype

2

Mendelian  
patterns

3

Punnett-  
square  
prediction

4

Beyond  
simple  
dominance

5

Pedigrees  
and  
probability

**Lab evidence**

lab-09\_inheritance-genetics.md

## INHERITANCE GENETICS

# Learning objectives

- Distinguish gene, allele, genotype, and phenotype.
- Use segregation to explain monohybrid inheritance.
- Build and interpret Punnett-square probabilities.
- Recognize inheritance patterns beyond complete dominance.
- Use pedigree evidence to reason about traits.

**OBJECTIVE 1**

Distinguish gene, allele, genotype, and phenotype.

**OBJECTIVE 2**

Use segregation to explain monohybrid inheritance.

**OBJECTIVE 3**

Build and interpret Punnett-square probabilities.

**OBJECTIVE 4**

Recognize inheritance patterns beyond complete dominance.

**OBJECTIVE 5**

Use pedigree evidence to reason about traits.

## INHERITANCE GENETICS

# Topic sequence

- Alleles and genotype: Alleles are gene variants that contribute to genotype and phenotype.
- Mendelian patterns: Mendelian inheritance predicts patterns when alleles segregate into gametes.
- Punnett-square prediction: Punnett squares model probability, not guaranteed outcomes.
- Beyond simple dominance: Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.
- Pedigrees and probability: Pedigrees use family evidence to infer inheritance patterns.

01

**Alleles and genotype**

Alleles are gene variants that contribute to genotype and phenotype.

02

**Mendelian patterns**

Mendelian inheritance predicts patterns when alleles segregate into gametes.

03

**Punnett-square prediction**

Punnett squares model probability, not guaranteed outcomes.

04

**Beyond simple dominance**

Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.

05

**Pedigrees and probability**

Pedigrees use family evidence to infer inheritance patterns.

## INHERITANCE GENETICS

# Module 09: Inheritance Genetics Evidence Map

- Module focus: Inheritance Genetics
- Central claim: Inheritance Genetics explains allele segregation with probability, phenotype prediction, and pedigree evidence.
- Key nodes to connect: Alleles and genotype, Mendelian patterns, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-09\_inheritance-genetics.md supplies an evidence surface for the map.

**Teaching note:** Ask students to explain one edge in their own words before moving on.

## INHERITANCE GENETICS

# Module 09: Inheritance Genetics Reasoning Sequence

- Module focus: Inheritance Genetics
- Trace the model: Alleles and genotype -> Mendelian patterns -> Punnett-square prediction -> Beyond simple dominance
- Inputs become outputs: Alleles and genotype, Mendelian patterns -> Distinguish gene, allele, genotype, and phenotype., Use segregation to explain monohybrid inheritance.
- Feedback check: lab-09\_inheritance-genetics.md evidence checks whether students can explain punnett-square prediction.
- Lab connection: lab-09\_inheritance-

**Teaching note:** Emphasize where the process can fail, slow down, or be revised by evidence.

## INHERITANCE GENETICS

# Terms as evidence handles

- Gene: DNA sequence contributing to a trait or functional product.
- Allele: Variant form of a gene.
- Genotype: Allele combination of an organism.
- Phenotype: Observable trait or characteristic.
- Dominant: Allele effect visible with one copy in simple dominance.
- Recessive: Allele effect masked by dominant allele in heterozygotes.

**GENE**

DNA sequence contributing to a trait or functional product.

**ALLELE**

Variant form of a gene.

**GENOTYPE**

Allele combination of an organism.

**PHENOTYPE**

Observable trait or characteristic.

**DOMINANT**

Allele effect visible with one copy in simple dominance.

**RECESSIVE**

Allele effect masked by dominant allele in heterozygotes.

## INHERITANCE GENETICS

# Lab connection

- Lab file: lab-09\_inheritance-genetics.md
- Lab evidence should test or illustrate: Punnett-square prediction
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**

Alleles and genotype

**EVIDENCE**

lab-09\_inheritance-genetics.md

**CLAIM**

Distinguish gene, allele, genotype, and phenotype.

## INHERITANCE GENETICS

# Contrast check

- Do not stop at: Alleles are gene variants that contribute to genotype and phenotype.
- Stronger explanation: Pedigrees use family evidence to infer inheritance patterns.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

Alleles are gene variants that contribute to genotype and phenotype.

**DEEPER**

Pedigrees use family evidence to infer inheritance patterns.



## INHERITANCE GENETICS

# Module 09: Inheritance Genetics Retrieval and Lab Check

- Module focus: Inheritance Genetics
- Retrieval prompt: How are genes and alleles related?
- Answer check: Distinguish gene, allele, genotype, and phenotype.
- Required terms: Gene, Allele, Genotype
- Lab connection: lab-09\_inheritance-genetics.md; lab-09\_inheritance-genetics.md supplies evidence for crosses, pedigrees, and probability patterns in inheritance.

**Teaching note:** Pause here. Students answer first without notes, then compare to the checks.

## INHERITANCE GENETICS

# Practice quiz bridge

- 1. An organism's genotype is its:
- 2. Two heterozygotes are crossed ( $Aa \times Aa$ ). What fraction of offspring are expected to show the recessive trait?
- 3. A heterozygote ( $Aa$ ) shows the dominant trait because:
- 4. What is a pedigree used for?

**QUESTION 1****Key: C**

An organism's genotype is its:

**QUESTION 2****Key: A**

Two heterozygotes are crossed ( $Aa \times Aa$ ). What fraction of offspring are expected to show the recessive trait?

**QUESTION 3****Key: D**

A heterozygote ( $Aa$ ) shows the dominant trait because:

**QUESTION 4****Key: B**

What is a pedigree used for?

## INHERITANCE GENETICS

# Synthesis and exit ticket

- Exit claim: explain alleles and genotype using evidence from lab-09\_inheritance-genetics.md.
- Must include: Gene, Allele, Genotype.
- Revision prompt: what evidence would change your explanation?

**CLAIM**

Alleles and genotype

**EVIDENCE**

lab-09\_inheritance-genetics.md

**REVISION**

What would change your mind?